

Tran Duc Trung

Phone: +(84) 325647395

Email: trung1803lucky@gmail.com

[Github](#) | [Linkedin](#)

1. Goals and objectives

I am seeking an opportunity in an organization that offers me a chance to learn, work, and contribute to the company's success. Big data and AI are my interests. My goal is to master the complexities of data engineering, while also expanding my knowledge and skills in AI and ML.

2. Education

VNU-HCM University of Science

2021 – 2025 (Expected Graduation)

- Bachelor of Data Science (GPA: 3.54/4.0)

English Certificate (VStep - B2)

August. 2024

3. Experience

Gene Solutions - Data Engineer Intern

Dec.2024 - Apr.2025

- Motivation: From fungal data in NCBI PubMed and patient data in the database, build a genomic pipeline to extract insights from the data and determine if there is any relationship between a fungal type and a cancer type.
- Tasks:
 - Collect and set up resources, design a set of genome references for fungi species
 - Researching each step of the bioinformatic pipeline from QC, read mapping, to taxonomic profiling, Kraken and BWA algorithms.
 - Building pipelines, compiling results for downstream analysis
 - Investigate the association between cancer/healthy samples and their fungal profiles.
 - Read papers and conduct research on AI in Bioinformatics.
- Technologies: Docker, Nextflow orchestrator

TMA Solutions - AI Intern

Jul-Sep. 2024

- Motivation: Develop a robust deepfake voice detection system with limited resources.
- Task:
 - Researched audio processing techniques (wav2vec, BEATs) and applied pretrained models for deepfake voice recognition based on "Training-Free Deepfake Voice Recognition" paper.
 - Vectorized reference audio dataset using pretrained models and stored in Qdrant vector database.
 - Vectorized input dataset, compare input audio vectors with references for identity verification and deepfake detection.
 - Optimized performance by testing pretrained models, similarity score thresholds, and reference set sizes to determine the best configuration for accuracy and efficiency.
- Technologies: Deep learning for audio, Qdrant

4. Projects

Data Engineering:

ETL Data Pipeline For Trip Record

Oct – Dec. 2023

- Motivation: Build an ETL data pipeline with the TLC Trip Record Data. Helping taxi businesses acquire sufficiently good and clean data before conducting statistical analysis or building models for prediction, thus deriving insights to enhance taxi services based on observed patterns within the data.
- Github: [NYC-TripRecord](#) with [#Demo](#)
- Tasks:
 - Ingest data into the database.
 - Develop end-to-end data platform by building ETL pipeline, batch processing with Apache Spark.
 - Transfer data to a data warehouse.
 - Make some analysts and demo data with a web app written by Streamlit.
- Technologies: Docker, Dagster, Apache Spark, MySQL, MinIO, PostgreSQL, Streamlit.

AI / Machine Learning:

Visual Question Answering in Medical VQA-RAD data

June – July. 2024

- Motivation: Developing a medical VQA system aims to enhance diagnostic accuracy and efficiency by providing instant, precise answers to questions based on medical images, which supports healthcare
- Github: [Project-NLP-VQA](#)
- Task:
 - Applied MUMC (Multi-Modal Utility Contrastive Learning) for VQA-Rad. Focusing on how it approaches understanding relationships between different modalities during pre-training, MUMC aims to unify different contrastive learning approaches across modalities
 - Read paper, explore contrastive learning, train both pre-training phase (as Image Captioning) and fine tuning phase (as VQA).
- Technologies: Pytorch, NLP knowledge

5. Skills & Coursework

- **Programming Languages:** Python, C/C++.
- **My tech stack:** Basic Python libraries, Scikit-learn, Pytorch, Spark, Hadoop, Dagster/Airflow, dbt, MinIO, Docker, Linux (Ubuntu), FastAPI.
- **Database:** SQL (PostgreSQL, MySQL), NoSQL (MongoDB).
- **Relevant School Coursework:** DSA, OOP, Databases, Database Management Systems, Intro to AI, Data Mining, Machine Learning, Deep Learning, Pattern Recognition, NLP, BigData.

6. Activities & Certifications

- [Fundamental Data Engineering](#) #AIDE Institute
- [VIASM The Summer School In Data Science 2023](#) #VNU-HCMUS
- [HackerRank SQL \(Basic to Advanced\) Skills Certifications](#) #HackerRank
- [First Prize Excellent Student At Math - City Level](#) #Mar 2021
- [IBM Data Engineering](#) #Coursera